Allison Newman

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Education

Cornell University, *Bachelor of Science, Mechanical Engineering* **GPA: 3.93 | Aug. 2024 - Exp. Dec. 2027**
Cornell University, *Masters of Engineering, Aerospace Engineering* **Jan. 2028 - May 2028**
Relevant courses: Statics, Mechanics of Materials, Dynamics, Fluid Mechanics, Intro to Aeronautics, Spaceflight Mechanics, Thermodynamics

Experiences

Lockheed Martin RMS–Sikorsky, *Assembly & Flight Operations, Manufacturing Eng. Intern* **May - Jul. 2026**
- Simulated and modeled gas-purging process using inert gas for AFSD with STAR-CCM+ and Ansys to demonstrate inefficiencies in current setup and suggest improvements for future systems. Familiarized myself with steady, transient and turbulent models; implemented passive scalar models.
- Sliced, printed, and packaged 26 to-scale parts of the modernized Black Hawk nose for a trade-show in under 2 weeks utilizing 3 F900 printers.
- Designed structural fixtures to stamp part-numbers onto welding substrates, remove laser powderbed fusion parts, and deposit welding stock.
- Conceptualized propulsion system for hybrid VTOL aircraft capable of providing 18HP and hovering for 20 minutes; awarded 2nd place. Performed trade study on motor & engine. Chose rotor and ESC to accompany motors; created test & safety guide for ground and flight tests.

Cornell University Unmanned Air Systems (CUAir), *Structures and Payloads Engineer* **Nov. 2024 - Present**
- Dedicating 9+ hours weekly to help design, manufacture, and test a carbon-fiber-composite search-and-rescue autonomous aircraft as part of the CUASC competition. Performed carbon-fiber layups of fuselage and bulkhead; weight-optimizing systems by 40% to fit within 35lb restraint.
- Placed second in flight and design at CUASC UAV 2025 international competition. Team routinely places first in innovation at SUAS UAV comp.
- Prototyping a novel plane-recovery system to prevent timeline setbacks of crashes. Weighing traditional ballistic launch vs. pneumatic systems.

Cornell University Intro Fluid Mechanics, *Undergraduate Teaching Assistant* **Aug. 2026 - Present**
- Demonstrating fluid concepts using Schlieren imaging, forklift hydraulics, and engine dissections to 3 classes of 20 third-year students per week.

Academic Excellence Workshop, *Facilitator for MATH2940 - Linear Algebra* **Aug. 2025 - May 2026**
- Curating 2-hour weekly workshops with co-facilitator to reinforce key Linear Algebra principles for undergraduate students; drafting, proofreading, and testing worksheets and games to engage students while highlighting key concepts, totaling 4+ hours of prep weekly.
- Working closely with Linear Algebra TAs and Office of Inclusive Excellence to establish a challenging and reflective workshop to improve students' understanding of difficult concepts by attending biweekly training to reinforce active learning principles built off of group collaboration.

Projects

Tilt Rotor Mechanism for 35lb eVTOL Aircraft, *CUAir Project*
- Designed and sized a tilt-rotor mechanism providing 45lbs of horizontal thrust at 25m/s, enabling high-speed flight mode transitions at 180 deg./sec. while maintaining minimum factors of safety (FOS) of 1.5 against horizontal thrust-to-weight ratio and 1.4 against bracket yield.
- Calculated load paths and critical stresses using first-principles statics and dynamics to evaluate boom, clevis, and gear geometry for bending, bearing, buckling, and yield margins.
- Validated stiffness and deflection limits via Ansys static structural finite element analysis (FEA) across the booms, brackets, gears, and bearings. Evaluated linkage vs. gear-driven actuation in SolidWorks to optimize rotation mechanics.
- Developed coupled thrust-and-rotation test protocols to validate kinematic performance, structural integrity, and operational safety limits.
- Performed crash analysis to identify load to induce yield on rotating bracket (50 Nm) and gear demeshing. To address deformation, designed a steel retaining plate to absorb load, increasing rigidity by ~8 times to prevent gears from demeshing.

Object Collecting Robot, *Mechatronics Project*
- Ranked 3rd out of 60 teams for collecting the most blocks against another robot opponent; collected 15 blocks/ min. with a custom mechanism.
- Designed in SolidWorks using COTS + 9 bespoke parts to be fabricated out of acrylic; assembled hands-on with laser-cutter, dremel, and adhesive.
- Programmed Arduino using C++ and digital registers. Utilized 3 servos and 2 QTIs + color sensor for board localization/ border detection.

Clubs and Volunteering

Society of Women Engineers, *Communications Codirector* **Sep. 2024 - Present**
- Contacted over 20 professors at Cornell University to secure a diverse group of mentors for the annual student-faculty dinner with 30+ attendees.

Skills and Certifications

Technical: SolidWorks (Sheet Metal), CATIA, 3DX, Ansys FEA (Static Structural), STAR-CCM+ CFD, GD&T, DFMA, Hand Calculations (Statics/Dynamics), Carbon Fiber Composite Layups & Testing, FDM 3D Printing (Insight, GrabCAD), Machining (Mill, Lathe, AFSD Welding), MATLAB, Arduino (C++), Java
Operations & Safety: Solumina, Confluence, Project Management, OSHA Safety, ESH, FOD Awareness